MATH100 IMPORTANT DEFINITIONS (Final term)

1) Define a set?
Definition: A **set** is an *unordered* collection of *distinct* objects. Objects in the collection are called **elements** of the set.

2) What is roster method of sets?
The **roster method** of specifying a set consists of surrounding the collection of elements with braces. For example the set of counting numbers from 1 to 5 would be written as {1, 2, 3, 4, 5}.

3) Define null set?
Definition: The set with no elements is called the **empty set** or the **null set.**

4) Define universal set?
Definition: The **universal set** is the set of all things pertinent to a given discussion and is designated by the symbol *U*

5) Define subset?
Definition: The set A is a **subset** of the set B, denoted A B, if every element of A is an element of B.

6) Define equal set?
Definition: Two sets A and B are **equal** if A B and B A. If two sets A and B are equal we write A = B to designate that relationship.

7) Define intersection of sets?
Definition: The **intersection** of two sets A and B is the set containing those elements which are elements of A **and** elements of B. We write A B to denote A Intersection B. Example: If A = {3, 4, 6, 8} and B = { 1, 2, 3, 5, 6} then A B = {3, 6}

8) Define union of sets?
Definition: The **union** of two sets A and B is the set containing those elements which are elements of A **or** elements of B. We write A B to denote A Union B. Example: If A = {3, 4, 6} and B = { 1, 2, 3, 5, 6} then A B = {1, 2, 3, 4, 5, 6}.

9) What are algebraic properties of sets? Or difference between commutative assosiative and distributive law?
Commutative: Union and intersection are **commutative** operations. In other words, **A** B = B A and A ∩ B = B ∩ A
Assosiative: Union and intersection are **associative** operations. In other words, **(A** B) C = A (B C) and (A ∩ B) ∩ C = B ∩ (A ∩ C)
Distributive: Union and Intersection are **distributive** with respect to each other. In other words **A** ∩ **(B** C)= (A ∩ B) (A ∩ C) and A (B ∩ C)= (A B) ∩ (A C)

10) Define cardinality? With two types?
Definition: **Cardinality** refers to the number of elements in a set
A *finite* set has a countable number of elements
An *infinite* set has at least as many elements as the set of *natural numbers*

11) Define complex number?
Definition: Numbers of the form *a* + *bi* are called **complex numbers.**

a is the **real** part and *b* is the **imaginary** part. The set of complex numbers is denoted by **C**

12) Defineabsolute value?

Definition: The **absolute value** or **modulus** of a complex number is the distance the complex number is from the origin on the complex plane.

13) Define relation?

Definition: A **mapping** between two sets A and B is simply a rule for relating elements of one set to the other. A mapping is also called a **relation**.

14) Define domain and range ?

Definition: The set consisting of members of the pre-image or inputs of a function is called its **domain.** For a given domain the set of possible outcomes or images of a function is called its **range**.

15) Define even and odd function?

Definition: A function is called an **even function** if its graph is symmetric with respect to the vertical axis, and it is called an **odd function** if its graph is symmetric with respect to the origin.

16) Define quadratic function?

Definition: A function of the type $y = ax2 + bx + c$ where *a, b,* and *c* are called the coefficients, is called a **quadratic function**.

17) Define a matrix?

Definition: A **matrix** is a rectangular arrangement of numbers in rows and columns. The **order** of a matrix is the number of the rows and columns. The **entries** are the numbers in the matrix.

18) Define identity matrix?

Definition: A Square matrix with ones on the diagonal and zeros elsewhere is called an **identity** matrix. It is denoted by *I.*

19) When a matrix has echelon form?

A matrix is in **echelon form** if it has the following properties
Every non-zero row begins with a 1 (called a **leading** 1)
Every leading one in a lower row is further to the right of the leading one above it.
If there are zero rows, they are at the end of the matrix

20) When a matrix has refuced echelon form?

A matrix is in **reduced echelon form** if in addition to the above three properties it also has the following property:
Every other entry in a column containing a leading one is zero
Methods for finding Solutions of Equations:
Using Row Operations: Recall that when we are solving simultaneous equations, the system of equations remains unchanged if we perform the following operations:
Multiply an equation by a non-zero constants
Add a multiple of one equation to another equation
Interchange two equations.

21) Define adjoin or adjugate?

Definition: Given a matrix *A*, calculate all the cofactors of *A*. We then form the matrix (of the cofactors. The **Adjoint** or **Adjugate** of *A* is the transpose of the matrix of the cofactors.

22) Define sequence and terms?
Definition: Rows of numbers are called **sequences**, and the separate numbers are called **terms** of the sequence.

23) Define arithmetic sequences?
Definition: An **Arithmetic Sequence** (or **Arithmetic Progression**) is a sequence in which each term after the first term is found by adding a constant, called the common difference (*d*), to the previous term

24) Define geometric sequences?
Definition: A sequence in which each term after the first is found by multiplying the previous term by a constant value called the common ratio, is called a **Geometric Sequence** (or **Geometric Progression**). The formula for finding any term of a geometric sequence is *an = arn-1*

25) Define geometrics series?
Definition**:** A **Geometric Series** is the sum of the terms in a arithmetic sequence. The formula for fining the sum of the first n terms of a geometric sequence is given by Sn=a(1-rn)/1-

26) Define convergent and divergent?
Definition: If a sequence of numbers approaches (or converges) to a finite number, we say that the sequence is **convergent.** If a sequence does not converge to a finite number it is called divergent.

27) What is multiplication principal?
Multiplication Principle: If two operations A and B are performed in order, with *n* possible outcomes for A and possible outcomes for B, then there are n x m possible combined outcomes of the first operation followed by the second.

28) Describe Pascal's Triangle?
Expression= Coefficients
(x + y)1 = x + y= 1 1
(*x* + *y*)2 = *x*2 + 2*xy* + *y*=21 2 1
(*x* + *y*)3 = *x*3 + 3*x*2*y* + 3*xy*2 + *y*3= 1 3 3 1
(*x* + *y*)4 = *x*4 + 4*x*3*y* + 6*x*2*y*2 + 4*xy*3 + *y*=4 1 4 6 4 1
(*x* + *y*)5 = *x*5 + 5*x*4*y* + 10*x*3*y*2 + 10*x*2*y*3 + 5*xy*4 + *y*5 =1 5 10 10 5 1

29) What are Properties of Gradient?
The bigger the gradient's magnitude is, the steeper the line segment.
Negative gradient means line is facing downwards.
Positive gradient means the line is facing upwards.
The slope gives the average rate of change in y per unit change in x, where the value of y depends on x.
Two line segments that are parallel will have the same slope.

30) What are Properties of the Cosine Function?
The cosine function has "period" 360*o* as it repeats itself after each revolution of 360*o*
Cos(-) = Cos as the x-coordinate of P doesn't change when we reflect across the x-axis.
Cos (180 -) = - Cos as the x-coordinate changes signs when reflected across the y-axis.
Cos (- 180) = - Cos as the x-coordinate changes signs when reflected across the origin.

Cos is positive in the first and the fourth quadrant (as the x coordinate of P is positive), and negative in the second and the third quadrant as the x-axis is negative there.
The range of the cosine function is between -1 and 1. The maximum value of 1 is taken when = 0*o*, ±360*o*, ±720*o*, …, and the minimum value of -1 is at = ±180*o*, ±540*o*, …,

31) Define periodicfunction and period?
The functions with the property that they keep repeating themselves are called **periodic functions**. The smallest interval for which the function repeats itself is called its **period**.

32) What are Properties of the Sine Function?
Sin(-) = -Sin as the y-coordinate of P changes sign when we reflect across the x-axis.
Sin is positive in the first and the second quadrant (as the y coordinate of P is positive), and negative in
the third and the fourth quadrant as the y coordinate is negative there.
Sin(180 -) = Sin because as we reflect across the y-axis the y-coordinate doesn't change.
Sin(- 180) = - Sin as the y-coordinate changes signs when reflected across the origin.
Like the cosine function, the sine function is also periodic, with period 360 degrees, and range between -1 and 1.

33) What are Properties of the Tangent Function?
Tan(-) = -Tan as the y-coordinate of P changes sign when we reflect across the x-axis but the x-coordinate doesn't change sign.
Tan is positive in the first and the third quadrant (as the x and y coordinates of P have the same signs in these quadrants), and negative in the second and the fourth quadrant as the x and y coordinates have opposite signs in these quadrants.
Tan(180 -) = -Tan as the x-coordinate of P changes sign when we reflect across the y-axis but the y-coordinate doesn't change sign
The domain of tan does **not** include the angles for which x is 0, namely, for = ±90, ±270, …
Like the cosine and sine functions, the tangent function is also periodic, but its period is 180. i.e. tan(+ 180) = tan and tan(- 180) = tan

34) Define amplitude?
Definition: The **Amplitude** of a function is the height from the mean (or the rest) value of the function to its maximum or minimum value.

35) Define inverse sin function?
Definition: The **inverse sine function** is defined by *y* = arcsin *x* if and only if sin *y* = *x*. The domain of *y* = arcsin *x* is [–1, 1]. The range of *y* = arcsin *x* is [–/ , /2].

36) Define inverse cosin function?
Definition: The **inverse cosine function** is defined by *y* = arccos if and only if cos *y* = *x*. The domain of *y* = arccos *x* is [–1, 1]. The range of *y* = arccos *x* is [0,].

37) Define inverse tangent function?
Definition: The **inverse tangent function** is defined by *y* = arctan *x* if and only if tan *y* = *x*. The domain of *y* = arctan *x* is . The range of *y* = arctan *x* is [–/2, /2].

38) Define identity and conditional equation?
Definition: Two functions *f* and *g* are said to be **identically equal** if *f(x)* = *g(x)* for **every value** of x for which both functions are defined. Such an equation is referred to as an **identity**. An equation that is not an identity is called a **conditional equation.**

39) Define statistics ?
Definition: Methods of collection, organization and analysis of numerical information are collectively called **statistics.**
40) Define variable and data?
Pieces of numerical and non-numerical information are called **data.** In order to collect data, you need to observe or measure some property; this property is called a **variable**.

41) Define quantitative and qualitative variables?
Definition: A variable is **qualitative** if it is not possible for it to take a numerical value. A variable is **quantitative** if it can take a numerical value. A quantitative variable which can take any value in a given range is called a **continuous** variable. A quantitative variable which has clear steps between its possible values is called a **discrete** variable.

42) Define Descriptive Statistics ?
comprises those methods concerned with collection and describing a set of data so as to yield meaningful information is called descriptive statistics.

43) Define inferential statistics?
Inferential Statistics comprises those methods concerned with analysis of a subset of data leading to predictions or inferences about the entire set of data.

44) Define population?
Definitions: **A population** is defined as the set of all possible members of a stated group. A cross-section of the returns of all of the stocks traded on the New York Stock Exchange (NYSE) is an example of a population.

45) Define sample?
Definition**: A sample** is defined as a subset of the population of interest. Once a population has been defined, a sample can be drawn from the population, and the sample's characteristics can be used to describe the population as a whole.

46) Define perimeter?
Definition: A measure used to describe a characteristic of a population is referred to as a **parameter**.

47) What is Nominal scale?
Observations are classified or counted with *no particular order*. It consists of assigning items to groups or categories. *No quantitative information* is conveyed and *no ordering* of the items is implied. Nominal scales are therefore *qualitative* rather than quantitative.
Religious preference, race, and gender are all examples of nominal scales.

48) What is Ordinal scale?
All observations are placed into separate categories and the *categories are placed in order with respect to some characteristic*. Differences between values makes no sense.
Political parties on left to right spectrum given labels 0, 1, 2; restaurant ratings, etc, are examples of ordinal scales.

49) Define Interval scale?
This scale provides *ranking* and assurance that differences between scale values are equal.
Difference makes sense, but ratio doesn't; and there is *no natural zero*. temperature (C,F) and dates are examples of interval scale

50) What is Ratio scale?

These represent the strongest level of measurement. In addition to providing ranking and equal differences between scale values, ratio scales have a *true zero point as the origin*. Height, weight, age and length are all examples of ratio scale.

51) What is frequency disribution?

Definition: A **frequency distribution** is a tabular presentation of statistical data that aids the analysis of large data sets. Frequency distributions summarize statistical data by assigning it to specified groups, or intervals.

52) Define relative distribution?

Definition: **Relative frequency** is calculated by dividing the frequency of each interval by the total number of observations. Simply, relative frequency is the percentage of total observations falling within each interval.

53) Definecommulative frequency?

Definition**: Cumulative** Frequencyis calculated by summing the frequencies starting at the lowest interval and progressing through the highest. Cumulative frequency for any given interval is the sum of the frequencies up to and including the given interval.

54) What is bar chart?

Definition: A **Bar chart** graphically represents the data sets by representing the frequencies as heights of bars.

55) What is histogram?

Definition: A Bar chart which represents continuous data is called a **histogram** if

The bars have no spaces between them (though there may be bars with zero height, which could look like spaces).

The area of each bar is proportional to the frequency.

If all the bars have the same width, then the height is proportional to the frequency.

56) Define frequency density?

Definition: **Frequency Density** is defined as the ratio between the frequency of a class and the class width. i.e. Frequency Density = Frequency/Class Width

57) Define arithmetic mean?

Definition: The **Arithmetic Mean** is the sum of the observation values divided by the number of observations. It is the most widely used measure of central tendency, and is the *only measure* where the *sum of the deviations* of each value from the mean is always *zero*.The formula for calculating the arithmetic mean of *n* values is: 1 n i i X X n ☐ ☐ ☐

58) Define probability space ?

Definition: A **Probability Space** or **Sample Space** is the set of all the possible outcomes in an experiment

59) What are Characteristics of a Probability Distribution?

The probability of any event A lies between 0 and 1, i.e. 0()PA

The sum of probabilities of all the mutually exclusive and exhaustive events in a probability distribution equals 1.

60) What is the probability of selecting a truck at random that has either air bags or bucket seats?

Solution: The addition rule for probabilities is used to determine the probability of at least one event among two or more events

occurring. The probability of each event is added and the joint probability (if the events are not mutually exclusive) is subtracted to arrive at the solution. P(air bags or bucket seats) = P(air bags) + P(bucket seats) - P(air bags and bucket seats) = (125/220) + (110/220) - (75/220) = 0.57 + 0.50 0.34 = 0.73 or 73 percent.

Alternative: 1 - P(no airbag and no bucket seats) = 1 - (60/220) = 73%